During the development of this prototype, I followed a top-down approach, starting from core systems and progressively refining smaller features. I began by implementing a basic character movement system, iterating on it to improve responsiveness and animation fluidity. I used Unity's new Input System to ensure scalability and easier future integration, such as adding controller or joystick support.

Next, I implemented a third-person camera system designed for an RPG perspective, ensuring it would not clip through environment objects to maintain immersion and usability.

The interaction system was built using an interface-based approach (IInteractable), enabling flexible interaction with NPCs and world objects. This led into the inventory system, which supports adding, removing, dragging, swapping, and consuming items. When an item is dropped into a slot, the system checks for occupancy and either swaps items or assigns it to an empty slot. The design is extensible, allowing different item types such as potions, weapons, and quest items to define their own behaviors through a centralized controller.

After completing the inventory system, I implemented a save and load system to persist its state. This was done afterward to ensure a stable data structure for properly serializing and deserializing items, maintaining slot consistency across sessions.

For animations, I used a blend tree for smooth movement transitions. I also implemented a basic combat system with combo attacks using Scriptable Objects, combined with root motion for better animation alignment.

Considering the 48-hour timeframe, I delivered multiple functional and extensible systems, while documenting potential improvements and future extensions within the code.